

Bridge Day 5 STUDENT QUIZ

Compound Boolean Evidence and Complete Decisions

Learning sequence: Predict - Trace - Explain - Test - Interpret

Friday, July 10, 2026 • 20 points • target 16 minutes

Name		UIN		Section	
------	--	-----	--	---------	--

Assessment boundary and evidence source

Contract: 07a04826c975

Cumulative foundations: input conversion, comparisons, and/or/not, branch order, f-strings

Scaffold level: complete compound-condition program

Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.13, 18.14, 18.15, 18.16, 18.17.

Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Compare two Boolean groupings. - 5 points

```
trained = False
temperature = 22
override = True

ready = trained and 15 <= temperature <= 30 or override
safe = trained and (15 <= temperature <= 30 or override)
print(ready, safe)
```

Question: Evaluate each component and explain why the two stored results differ.

Item 2. Trace negation and a dependent readiness rule. - 5 points

```
flow_ok = True
label = "pending"
needs_review = not flow_ok or label != "approved"
ready = flow_ok and not needs_review
print(needs_review, ready)
```

Question: Evaluate each Boolean in dependency order and give exact output.

--

Complete program - 10 points

- Read flow as a float, approval as text, and sensor as text.
- Print invalid if flow is negative.
- Otherwise print ready only when flow is from 18 through 22 inclusive, approval is exactly yes, and sensor is exactly ready.
- Print hold for every other nonnegative case. Print exactly one word.

Program workspace**Test input, expected result, and why this test matters**